

# PEFT-Arena: Understanding Parameter-Efficient Finetuning from a Stability-Plasticity Perspective

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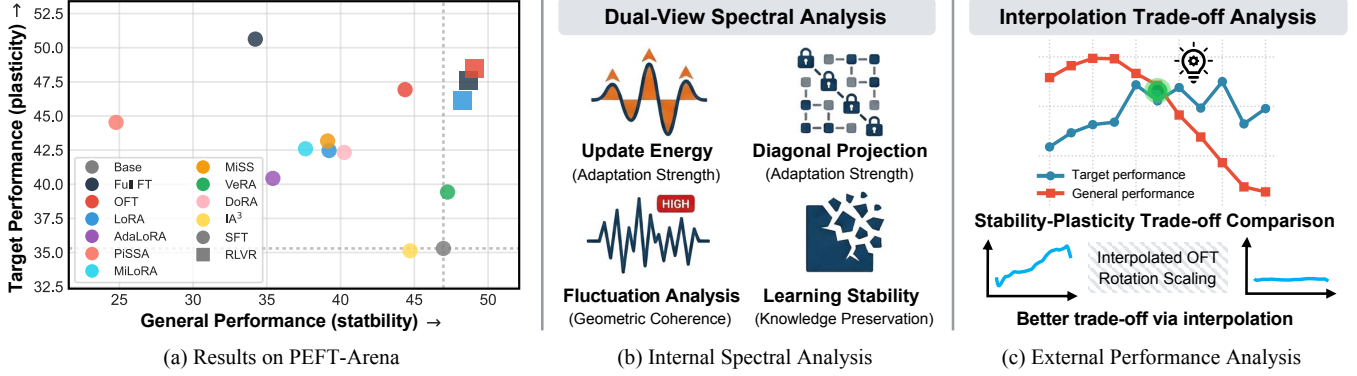


Figure 1: PEFT-Arena comprehensively evaluates the downstream task adaption and pretrained knowledge retention of PEFTs in LLM post-training. (a) Stability-plasticity trade-off on the downstream math task; (b) Internal analysis includes studying spectral geometry on weight matrices; (c) External analysis includes target and general performance evaluation and interpolation study.

## Abstract

Parameter-efficient finetuning (PEFT) has emerged as a practical solution for adapting large foundation models by updating only a small subset of parameters, and yet existing evaluations largely focus on downstream task performance while overlooking the preservation of pretrained capabilities. In this work, we argue that PEFT should be evaluated through the lens of the stability-plasticity dilemma, which characterizes the fundamental trade-off between efficient task adaptation (plasticity) and resistance to forgetting (stability). To this end, we introduce PEFT-Arena, a unified benchmark that jointly measures downstream performance and general capability retention. Our results show that all PEFT methods exhibit inherent stability-plasticity trade-offs and that different methods produce distinct trade-off patterns, indicating that neither metric alone is sufficient for evaluation. Besides external task-level assessment, we also propose to analyze the spectral geometry of weight updates to uncover the underlying mechanisms that govern the plasticity-stability trade-off. Our results show that PEFT methods achieving better trade-offs exhibit more structured and predictable spectral dynamics, highlighting spectral regularity as an intrinsic factor governing stability and as a guiding principle of the design of future PEFT algorithms. Inspired by the stability-plasticity trade-off, we exploit interpolation between the PEFT-tuned model and the base model. We find that such interpolated models often achieve a better trade-off than either model alone.

## CCS Concepts

• **Computing methodologies** → **Neural networks**; *Supervised learning*; *Reinforcement learning*; *Transfer learning*.

## Keywords

Large Language Models, Parameter-Efficient Finetuning, Orthogonal Finetuning, Catastrophic Forgetting, Spectral Preservation

## 1 Introduction

As foundation models continue to scale, finetuning all parameters becomes increasingly infeasible. To address this challenge, parameter-efficient finetuning (PEFT) has been proposed. PEFT aims to adapt large pretrained models to downstream tasks by modifying only a small number of parameters, or by introducing lightweight trainable components, while keeping the majority of the model weights fixed. By drastically reducing the number of trainable parameters, PEFT significantly lowers computational and memory costs, enabling efficient adaptation across diverse tasks without the need to retrain or store a full model for each task.

Although numerous PEFT methods [1, 3, 6, 7, 11, 13–16, 18, 22, 23, 27, 32–34, 39, 44, 46, 60, 61, 64, 69, 74, 76] have been proposed, their evaluation has largely been limited to downstream task performance. We argue that this metric alone can be misleading and insufficient for fairly assessing the effectiveness of PEFT methods. Developing more principled evaluation criteria is therefore crucial for advancing PEFT research and calls for greater attention from the community. To address this limitation, we draw inspiration from the stability-plasticity dilemma [49], a well-known challenge in both artificial and biological neural systems. This dilemma characterizes the fundamental trade-off between learning efficiency (*i.e.*, plasticity) and resistance to forgetting (*i.e.*, stability). Guided by this dilemma, we are interested in the following question:

*Which PEFT method can achieve the best trade-off between stability and plasticity trade-off?*

To address this question, we introduce **PEFT-Arena**, a comprehensive benchmark for evaluating PEFT methods. This new PEFT benchmark focuses on assessing the trade-off between the task adaptation effectiveness (*i.e.*, downstream task performance) and the preservation of pretrained capabilities (*i.e.*, general task performance). By providing a reliable and unified evaluation framework, PEFT-Arena enables effective and fair comparison across different PEFT methods. With PEFT-Arena, we make a few useful observations: (1) neither downstream target performance nor general task performance can serve as a reliable evaluation metric (both metrics are necessary); (2) the stability-plasticity trade-off holds for every PEFT method; (3) different PEFT methods yields different trade-off patterns. More importantly, we find that the stability-plasticity perspective calls into question many recent methodological advances in PEFT, suggesting that future PEFT algorithms should be explicitly designed for improving this trade-off.

While PEFT-Arena offers an effective external empirical measure of stability and plasticity, it remains unclear which internal mechanisms best govern the stability-plasticity trade-off in PEFT. In order to gain deeper insights to designing better PEFT method, we therefore seek to answer the following question:

*What intrinsic metrics can characterize a good PEFT method?*

To this end, we propose to analyze how the spectral geometry of weight matrices evolves under different PEFT methods. This motivates us to monitor the training dynamics of the singular values and singular vectors of weight matrices. We find that PEFT methods exhibiting a favorable stability-plasticity trade-off tend to be more geometrically principled, in the sense that their weight updates are more spectrally structured and the spectral deviation before and after post-training is more predictable. In general, our empirical study implies that the stability of PEFT largely depends on spectral regularity and consistency, whereas plasticity is more multifaceted and can be achieved through less structured updates.

The stability-plasticity trade-off can be partially characterized by the distance between the finetuned model and the base model. Guided by this insight, we exploit interpolation between these two models, yielding interpolated variants of PEFT methods. We find that these interpolated variants often achieve a substantially better trade-off than either the finetuned or the base model alone. In the light of this observation, we further explore *interpolated* and *geometry-aware* variants of Orthogonal Finetuning (OFT). In particular, we introduce **interpolated Orthogonal Finetuning (iOFT)**, which adjusts OFT updates based on their intrinsic geometric structure. Empirically, iOFT serves as a practical and relatively robust trade-off knob, and can improve the stability-plasticity frontier over the vanilla OFT checkpoint without requiring an exhaustive sweep over a global scaling coefficient. Our contributions are listed below:

- **External understanding of PEFT.** We introduce PEFT-Arena, a unified benchmark that evaluates PEFT via the stability-plasticity trade-off, enabling comparison beyond downstream accuracy.

- **Internal understanding of PEFT.** We characterize PEFT mechanisms by monitoring *spectral dynamics* (singular values/vectors) during post-training, showing that better stability is associated with more structured spectral deviations.
- **Interpolated variants.** We show that interpolating between the base and finetuned models can often improve the trade-off, and explore iOFT as a geometry-aware OFT variant that can robustly improve the trade-off frontier.

## 2 Related Work

**PEFT and subspace learning.** Parameter-efficient finetuning (PEFT) specializes large pretrained models by optimizing only a small subset of parameters, substantially reducing computational overhead while often achieving performance comparable to full finetuning [17]. Prior PEFT methods can be broadly grouped into three categories. (1) *Additive weight reparameterization.* A representative approach is Low-Rank Adaptation (LoRA) [17], which freezes pretrained weights and introduces a learnable low-rank update to enable efficient task adaptation with a small number of trainable parameters. Subsequent work extends LoRA along several complementary dimensions. Methods such as AdaLoRA [75] focus on adaptive rank allocation by dynamically re-allocating rank budgets during training, and DoRA [40] improves optimization by decoupling update magnitude from direction. Other approaches modify the update structure: MISS [26] replaces the two-factor decomposition with a single expanded low-rank matrix, and VeRA [30] attains high effective update rank through shared, frozen random projection matrices modulated by lightweight trainable scaling vectors. Another line of work explores initialization strategies informed by the spectral properties of pretrained weights. PiSSA [48] initializes low-rank factors using principal singular components, whereas MiLoRA [65] instead targets minor components by initializing updates on the minor singular subspace while freezing the principal components. (2) *Constrained weight updates.* Beyond low-rank additive updates, Orthogonal Finetuning (OFT) [41, 53, 54] constrains the learned transformations to be orthogonal. By preserving the spectral properties of the original weight matrices, such constraints act as an implicit regularizer during adaptation and provide an alternative form of structured parameter efficiency. (3) *Activation- and prompt-based adaptation.* Instead of directly modifying weight matrices, this line of work introduces auxiliary parameters to modulate intermediate representations. For instance, IA<sup>3</sup> learns per-channel scaling vectors that element-wise modulate key activations [39]. Similarly, Prompt Tuning [42] and Prefix-Tuning [33] prepend trainable continuous embeddings to the input layer or to hidden states across attention blocks, respectively. These methods facilitate task specialization by re-contextualizing frozen representations, effectively inducing localized shifts in the activation space.

**Continual learning.** Continual learning (CL) aims to incorporate new knowledge while preserving previously acquired capabilities, thereby mitigating catastrophic forgetting [66]. Prior CL approaches are commonly categorized into data-replay-based methods [2, 59, 63], which require access to historical training data, and data-free methods [8, 43, 52, 70], which avoid replay by relying instead on architectural constraints or regularization. Since storing or revisiting past data can be impractical due to privacy, storage,

or compliance constraints, we focus primarily on data-free CL in this paper. This paradigm operates directly in parameter space, bypassing the need for historical data. Standard approaches employ  $\ell_2$  regularization [28] or model merging [29, 38, 43, 70] to anchor finetuned parameters to the base model, thereby balancing stability and plasticity. For PEFT, O-LoRA [67] and InfLoRA [35] enforce orthogonality between sequential task-specific subspaces to mitigate forgetting. However, these methods often overlook the preservation of pretrained knowledge. KeepLoRA [45] constrains updates to the non-principal subspace of the original weights, protecting the principal components that encode general capabilities.

**Catastrophic Forgetting in LLMs.** While traditional CL primarily addresses multi-task sequential learning, recent focus has shifted toward catastrophic forgetting during the post-training of LLMs, particularly when finetuning on distributions that diverge significantly from the pre-training data [37, 56]. A prevailing hypothesis identifies the discrepancy between on-policy and off-policy distributions as a primary driver of forgetting. To bridge this gap, several data-centric strategies aim to simulate on-policy learning via reweighting. FLOW [56] prioritizes simpler samples to stabilize updates, while TALR [37] and EAFT [9] mitigate forgetting by attenuating learning rates or weights for tokens with high difficulty or low information entropy. Despite the effectiveness of these data-level interventions, a fundamental question remains: how do these stability-promoting effects manifest in the model’s underlying parameter space? Emerging evidence in reinforcement learning suggests that optimization stability is intrinsically linked to parameter-space dynamics. Specifically, the preferential updating of non-principal weight components to constrain parameter shifts [50, 78]. While preliminary studies have empirically observed that LoRA exhibits superior knowledge retention compared to full finetuning [4], it remains poorly understood how such spectral dynamics evolve under different PEFT parameterizations, or how they structurally give rise to the resulting stability-plasticity trade-off. To address this gap, we conduct a systematic evaluation of PEFT methods to identify configurations that yield an optimal trade-off.

## 3 The PEFT-Arena Benchmark

### 3.1 Experimental Setup

In PEFT-Arena, parameter-efficient finetuning methods are evaluated along two axes: (i) target-domain performance and (ii) general ability preservation. We use target-domain performance as a protocol for *plasticity*, and general-task performance (capability retention) as a protocol for *stability*. We conduct experiments on two target domains: mathematical reasoning and medical reasoning. We conducted supervised finetuning (SFT) on two 10k scale training datasets for distillation of long chain-of-thought reasoning ability. Unless otherwise stated, we report average accuracy within each domain.

*Target-domain benchmarks and evaluation.* **(1) Math.** We evaluate on Math-500 [36], AMC23, and AIME24. **(2) Medical.** We evaluate on a collection of medical reasoning and knowledge benchmarks, including MedMCQA [51], MedQA (USMLE) [24], PubMedQA [25], MMLU-Pro [68], GPQA (Medical) [55], Lancet, NEJM & MedBullets

problems [5], and MedXpertQA [79] following a dedicated dataset survey [19].

*General ability preservation.* To measure general ability preservation after adaptation, we evaluate on IFEval [77], NQ [31], BBH [62], covering instruction following, natural language understanding, general knowledge and general reasoning. We report the average score across these tasks as a protocol to assess model forgetting after fin-tuning. We follow an OpenCompass-style evaluation configuration with context length 1024, temperature  $T = 0.0$ , and one sample per query.

*Base models and adaptation methods.* We use Qwen2.5-7B [72] and Llama3.2-3B-Instruct [10] as pre-trained backbones to cover different scales and base/instruction-tuned settings. We compare full finetuning (Full FT) against a representative set of PEFT baselines (see section 2 for background). **Additive PEFT (LoRA family).** We include LoRA [17] and representative variants spanning rank allocation/parameterization/initialization: AdaLoRA [75], DoRA [40], MiSS [26], VeRA [30], PiSSA [48], and MiLoRA [65]. **Spectrum Preserving updates (OFT).** We include orthogonal finetuning (OFT) [54], which constrains updates to structured (approximately) orthogonal transformations. **Activation-based PEFT.** We include IA<sup>3</sup> [39], a lightweight method that adapts models via learned activation scaling.

*Training and optimization details.* We conduct supervised finetuning (SFT) in both target domains, using 50k filtered samples from OpenR1-Math-330k [20] for math and 23k samples from MedThink [12] for medical. We also include RLVR post-training results with GRPO [58] on a representative subset of methods for comparison. Full schedules and optimization details are provided in subsection A.1.

### 3.2 Main results

*Supervised FineTuning (SFT).* As shown in Table 1, compared with PEFT methods, full finetuning (Full FT) achieves the largest gains on target-domain tasks with its full parameter plasticity, but it also exhibits substantial forgetting on general abilities. For instance, on Qwen2.5-7B-base, Full FT improves Math Target by +15.33 (35.30→50.63) and Med Target by +7.27 (46.36→53.63), but drops General by −12.74 for Math finetuning and by −12.56 for Med Finetuning; on Llama3.2-3B-Instruct, it drops General by −30.73 for Med Finetuning. This highlights that our evaluation setting can jointly assess both plasticity and stability for finetuning methods.

Different PEFT methods further present distinct plasticity-stability trade-offs. In terms of plasticity, most PEFT methods (e.g., LoRA, AdaLoRA, MiSS, and OFT) can acquire downstream reasoning ability, and the gain tends to increase with trainable parameters (e.g., on Qwen, MiSS  $r8$  reaches +7.87 on Math Target, while MiSS  $r64$  reaches +11.63). Notably, SVD-based initialization methods such as PiSSA are unstable in our setting: although PiSSA  $r8$  improves Qwen Math Target by +9.23, it severely degrades Qwen General by −22.19 and Med Target by −20.19 (and on Llama, General drops by −45.95). This motivates us to analyze the spectrum-level properties of PEFT updates.

| Method                                              | Hy. Param | Tr. Param | Qwen2.5-7B-base |                |                |                | Tr. Param | Llama3.2-3B-Instruct |                |                |                |
|-----------------------------------------------------|-----------|-----------|-----------------|----------------|----------------|----------------|-----------|----------------------|----------------|----------------|----------------|
|                                                     |           | (Qwen)    | Math Target     | Math General   | Med Target     | Med General    | (Llama)   | Math Target          | Math General   | Med Target     | Med General    |
| Supervised FineTuning (SFT)                         |           |           |                 |                |                |                |           |                      |                |                |                |
| Base                                                | –         | 7.61B     | 35.30 (+0.00)   | 46.97 (+0.00)  | 46.36 (+0.00)  | 46.97 (+0.00)  | 3.21B     | 27.63 (+0.00)        | 53.03 (+0.00)  | 41.44 (+0.00)  | 56.76 (+0.00)  |
| Full FT                                             | –         | 6.53B     | 50.63 (+15.33)  | 34.22 (-12.74) | 53.63 (+7.27)  | 34.41 (-12.56) | 2.85B     | 33.90 (+6.27)        | 39.83 (-13.20) | 44.26 (+2.82)  | 26.03 (-30.73) |
| OFT                                                 | block 16  | 8.49M     | 42.33 (+7.03)   | 42.58 (-4.39)  | 46.17 (-0.19)  | 45.09 (-1.88)  | 7.08M     | 29.43 (+1.80)        | 41.08 (-11.95) | 39.22 (-2.22)  | 40.97 (-15.79) |
| OFT                                                 | block 32  | 17.55M    | 46.93 (+11.63)  | 44.37 (-2.60)  | 48.63 (+2.27)  | 42.40 (-4.57)  | 11.55M    | 30.60 (+2.97)        | 40.73 (-12.30) | 39.50 (-1.94)  | 40.50 (-16.26) |
| OFT                                                 | block 64  | 35.68M    | 46.23 (+10.93)  | 35.97 (-11.00) | 49.47 (+3.11)  | 39.11 (-7.86)  | 24.97M    | 29.30 (+1.67)        | 39.75 (-13.28) | 40.77 (-0.67)  | 37.70 (-19.06) |
| OFT                                                 | block 128 | 71.92M    | 47.77 (+12.47)  | 36.98 (-9.99)  | 52.40 (+6.04)  | 36.88 (-10.08) | 47.34M    | 32.23 (+4.60)        | 36.26 (-16.76) | 42.17 (+0.73)  | 34.26 (-22.50) |
| LoRA                                                | r4a8      | 10.09M    | 42.33 (+7.03)   | 41.66 (-5.31)  | 47.12 (+0.76)  | 36.42 (-10.55) | 6.9M      | 24.30 (-3.33)        | 35.79 (-17.23) | 36.92 (-4.52)  | 31.84 (-24.92) |
| LoRA                                                | r8a16     | 20.19M    | 42.47 (+7.17)   | 39.22 (-7.75)  | 47.91 (+1.55)  | 36.06 (-10.91) | 12.16M    | 24.07 (-3.56)        | 36.57 (-16.46) | 38.34 (-3.10)  | 27.99 (-28.77) |
| LoRA                                                | r16a32    | 40.37M    | 44.87 (+9.57)   | 34.91 (-12.06) | 47.86 (+1.51)  | 34.86 (-12.11) | 24.31M    | 24.97 (-2.66)        | 37.55 (-15.48) | 39.21 (-2.23)  | 29.18 (-27.58) |
| LoRA                                                | r32a64    | 80.74M    | 45.37 (+10.07)  | 38.21 (-8.76)  | 49.48 (+3.12)  | 35.50 (-11.47) | 48.63M    | 25.90 (-1.73)        | 37.20 (-15.83) | 39.33 (-2.11)  | 30.69 (-26.07) |
| AdaLoRA                                             | r8a16     | 30.28M    | 40.43 (+5.13)   | 35.41 (-11.56) | 45.22 (-1.13)  | 37.34 (-9.63)  | 18.24M    | 20.83 (-6.80)        | 34.53 (-18.49) | 37.11 (-4.33)  | 36.29 (-20.47) |
| PiSSA                                               | r8a16     | 20.19M    | 44.53 (+9.23)   | 24.78 (-22.19) | 26.16 (-20.19) | 18.05 (-28.92) | 12.16M    | 0.67 (-26.96)        | 7.08 (-45.95)  | 21.17 (-20.27) | 9.75 (-47.02)  |
| MiLoRA                                              | r8a16     | 20.19M    | 42.60 (+7.30)   | 37.62 (-9.35)  | 46.83 (+0.48)  | 35.88 (-11.09) | 12.16M    | 23.60 (-4.03)        | 35.59 (-17.44) | 37.64 (-3.81)  | 29.23 (-27.53) |
| MiSS                                                | r8        | 11.12M    | 43.17 (+7.87)   | 39.12 (-7.85)  | 48.75 (+2.40)  | 34.43 (-12.54) | 6.19M     | 23.37 (-4.26)        | 33.93 (-19.09) | 40.16 (-1.28)  | 31.71 (-25.05) |
| MiSS                                                | r64       | 89.00M    | 46.93 (+11.63)  | 32.77 (-14.20) | 51.90 (+5.54)  | 32.72 (-14.25) | 49.55M    | 28.63 (+1.00)        | 34.96 (-18.06) | 41.96 (+0.52)  | 27.07 (-29.69) |
| VeRA                                                | r256      | 1.44M     | 39.43 (+4.13)   | 47.25 (+0.38)  | 28.50 (-17.85) | 47.01 (+0.04)  | 0.82M     | 28.80 (+1.17)        | 46.79 (-6.23)  | 40.68 (-0.76)  | 48.94 (-7.82)  |
| DoRA                                                | r8a16     | 21.58M    | 42.33 (+7.03)   | 40.25 (-6.72)  | 48.04 (+1.69)  | 36.06 (-10.91) | 12.93M    | 23.83 (-3.80)        | 35.65 (-17.37) | 38.25 (-3.19)  | 27.53 (-29.23) |
| IA <sup>3</sup>                                     | –         | 1.82M     | 35.13 (-0.17)   | 44.71 (-2.26)  | 29.70 (-16.66) | 45.72 (-1.25)  | 0.92M     | 29.70 (+2.07)        | 45.72 (-7.30)  | 39.13 (-2.31)  | 45.67 (-11.09) |
| RLVR with Group Relative Policy Optimization (GRPO) |           |           |                 |                |                |                |           |                      |                |                |                |
| Full FT                                             | –         | 6.53B     | 47.57 (+12.27)  | 48.68 (+1.71)  | 46.24 (-0.11)  | 43.22 (-3.75)  | 2.85B     | 29.80 (+2.17)        | 52.20 (-0.82)  | 45.88 (+4.44)  | 51.81 (-4.95)  |
| OFT                                                 | block 32  | 8.49M     | 48.37 (+13.07)  | 48.90 (+1.93)  | 46.79 (+0.44)  | 47.24 (+0.27)  | 11.55M    | 29.97 (+2.34)        | 50.04 (-2.98)  | 44.99 (+3.55)  | 52.31 (-4.45)  |
| LoRA                                                | r8a16     | 20.19M    | 46.10 (+10.80)  | 48.27 (+1.30)  | 47.08 (+0.73)  | 42.80 (-4.17)  | 24.31M    | 28.83 (+1.20)        | 52.17 (-0.85)  | 46.24 (+4.80)  | 53.54 (-3.23)  |

**Table 1: Main benchmark results. For each domain, average task accuracy is reported as score (Higher is better). We also report the performance change relative to the corresponding base model in parentheses.**

In terms of stability, methods that better preserve geometric properties of model parameters generally show less forgetting. For example, on Qwen, OFT (block 32) achieves strong target gains with much smaller forgetting than Full FT (Math Target +11.63 with General -2.60; Med Target +2.27 with Med General -4.57). VeRA best preserves general abilities (Qwen Math/Med General +0.38/+0.04) but sacrifices target performance (e.g., Qwen Med Target -17.85), while IA<sup>3</sup> exhibits relatively small forgetting (Qwen Math/Med General -2.26/-1.25) but limited target gains. Overall, OFT provides the most favorable plasticity-stability trade-off among the compared PEFT methods.

**Reinforcement Learning with Verifiable Rewards (RLVR).** In contrast to SFT, RLVR post-training with GRPO exhibits fundamentally different characteristics. As shown in Table 1, Full FT, OFT, and LoRA all achieve substantial improvements on math target domain (Full FT +12.27, OFT +13.07, LoRA +10.80) while exhibiting no forgetting on general abilities. In fact, all methods show slight improvements on General (+1.71, +1.93, +1.30 respectively). This phenomenon benefits from the on-policy nature of RLVR: unlike SFT which directly imitates training data distributions that may deviate from the pre-trained model’s behavior, RLVR optimizes the policy based on its own generated outputs, naturally preserving the model’s original capabilities while reinforcing correct reasoning patterns. Furthermore, RLVR post-training imposes minimal requirements on parameter capacity[57] for PEFT methods. Both LoRA (r8) and OFT (block 32) achieve competitive performance with Full FT despite using orders of magnitude fewer trainable parameters, suggesting

that the reinforcement learning objective is more efficiently captured by low-rank or structured parameterizations compared to supervised distillation.

**Findings and takeaways.** Overall, single-point comparisons at the final checkpoint can obscure the full stability-plasticity frontier, motivating us to move beyond a single operating point and explicitly trace trade-off curves. Moreover, the strong stability of OFT suggests that update geometry is a key factor; in the next section we connect this behavior to spectrum-level retention/adaptation diagnostics. Finally, our results also hint at an overshoot effect under SFT—the final checkpoint can lie past the best trade-off point—which we later address through interpolation-based and geometry-aware post-hoc adjustment.

Next, we analyze PEFT model geometry to further explain the underlying causes of the plasticity-stability trade-off in SFT.

## 4 Spectral Analysis of PEFT Adaptation

Existing work has studied the gap between PEFT and full fine-tuning by analyzing the spectrum of weight updates. Building on this line, we present a *Dual-View Spectral Analysis of Weight Updates* that connects different PEFT mechanisms to the stability-plasticity trade-off through two decoupled dimensions: **Retention** (how much of the pre-trained structure persists) and **Adaptation** (how the update energy is distributed), with a specific focus on **spectral smoothness** (local fluctuation across singular-value indices).

#### 4.1 Dual-View Spectral Analysis of Updates

*View I: Diagonal projection on the pre-trained basis (Retention).* Let the pre-trained weight be decomposed as  $W_{\text{pre}} = U\Sigma V^\top$ . We measure how much the fine-tuned weight  $W_{\text{ft}}$  preserves the pre-trained singular alignment via the diagonal projection

$$P_{\text{diag}}(i) = u_i^\top W_{\text{ft}} v_i. \quad (1)$$

We report the absolute change  $|\Delta P_{\text{diag}}(i)|$  to quantify **component-wise retention**: inconsistent changes suggest stronger deviation from the pre-trained component, often correlates with higher risk of interfering with pre-trained knowledge structure.

*View II: Update energy spectrum (Adaptation).* To capture where the optimization injects energy, we analyze the weight update  $\Delta W = W_{\text{ft}} - W_{\text{pre}}$  along the pre-trained input directions  $v_i$ :

$$E_\Delta(i) = \|\Delta W v_i\|_2. \quad (2)$$

Compared to  $P_{\text{diag}}(i)$ ,  $E_\Delta(i)$  reflects a more complete notion of adaptation magnitude, since it aggregates both scaling changes and orthogonal rotations along the  $i$ -th latent direction.

*Spectral smoothness and fluctuation modes.* For both  $|\Delta P_{\text{diag}}(i)|$  and  $E_\Delta(i)$ , we compute a **fluctuation score** as the distance to a moving average along the singular-value index. This characterizes whether adjacent components are treated consistently (smooth) or in a more “spiky” manner (high fluctuation). We summarize the observed patterns into four modes, these modes are used to represent tendencies rather than hard categories:

**Mode A: Plasticity-dominated** The update energy varies smoothly across indices, while  $|\Delta P_{\text{diag}}(i)|$  is relatively jagged. This often suggests globally distributed updates that may affect the pre-trained alignment in a more index-local manner; a plausible explanation is that a larger fraction of learning happens off-diagonal (feature re-allocation) rather than by rescaling the same components.

**Mode B: Coherent retention, sparse adaptation**  $|\Delta P_{\text{diag}}(i)|$  is smooth, but  $E_\Delta(i)$  shows spikes at a small number of indices. This pattern is consistent with preserving the global pre-trained scaling structure while injecting strong, orthogonal updates into a few “latent slots” for task-specific knowledge.

**Mode C: Global systematic adaptation** Both views change gradually across indices, indicating high spectral coherence. This corresponds to a more systematic transformation (e.g., style/domain shift) where similar features are updated in a similar way, which is often associated with stable optimization behavior.

**Mode D: Stochastic degradation** Both metrics fluctuate strongly across indices, suggesting that adjacent components are treated inconsistently. This pattern tends to correlate with unstable fitting and a higher likelihood of forgetting, although the severity depends on data/scale/regularization.

#### 4.2 Interpreting PEFT Stability-Plasticity

We now apply the above framework to interpret how different finetuning mechanisms manifest the stability-plasticity trade-off under *SFT* and *RLVR* [58].

*Full fine-tuning.* On **SFT**, Full FT tends to exhibit **Mode A**-like behavior: the update energy spreads broadly (high effective rank), while the retention view shows more localized fluctuations across indices. This is consistent with the intuition that SFT often requires holistic adjustment across both principal (semantic core) and minor (details) components. On **RLVR**, in contrast, Full FT more often resembles **Mode C**: both retention and adaptation evolve more smoothly. This matches the on-policy nature of RLVR updates, which empirically aligns with less destructive interference and better generalization[78].

*Low-rank additive PEFT.* LoRA[17] and its variants (e.g., AdaLoRA[75], MISS[26]) approximate  $\Delta W$  with a low-rank additive matrix ( $BA$ ), which naturally concentrates update energy into a small set of directions (low effective rank). Under **SFT**, this approximation frequently shows a **Mode D** tendency: both views become spiky, suggesting a geometric mismatch between the low-rank additive subspace and the pre-trained spectral geometry (i.e., updates behave more like “shortcuts” that cut across the manifold and can amplify forgetting). Under **RLVR**, however, LoRA often shifts toward **Mode B**: retention remains relatively coherent, while adaptation is sparse and targeted. This indicates that although LoRA is less plastic than Full FT in parameter space, it can still inject task-specific knowledge efficiently in RLVR while preserving the pre-trained structure.

*Explicit subspace partitioning.* PiSSA[48] and MiLoRA[65] attempt to target specific spectral regions (e.g., principal vs. minor components). In our observations under **SFT**, they still tend to fall closer to **Mode D**. A likely reason is that SFT updates couple principal and minor subspaces rather than cleanly separating them: aggressively modifying the principal components may increase interference, while restricting updates to minor components can limit domain adaptation (consistent with the limited gains in Table 1).

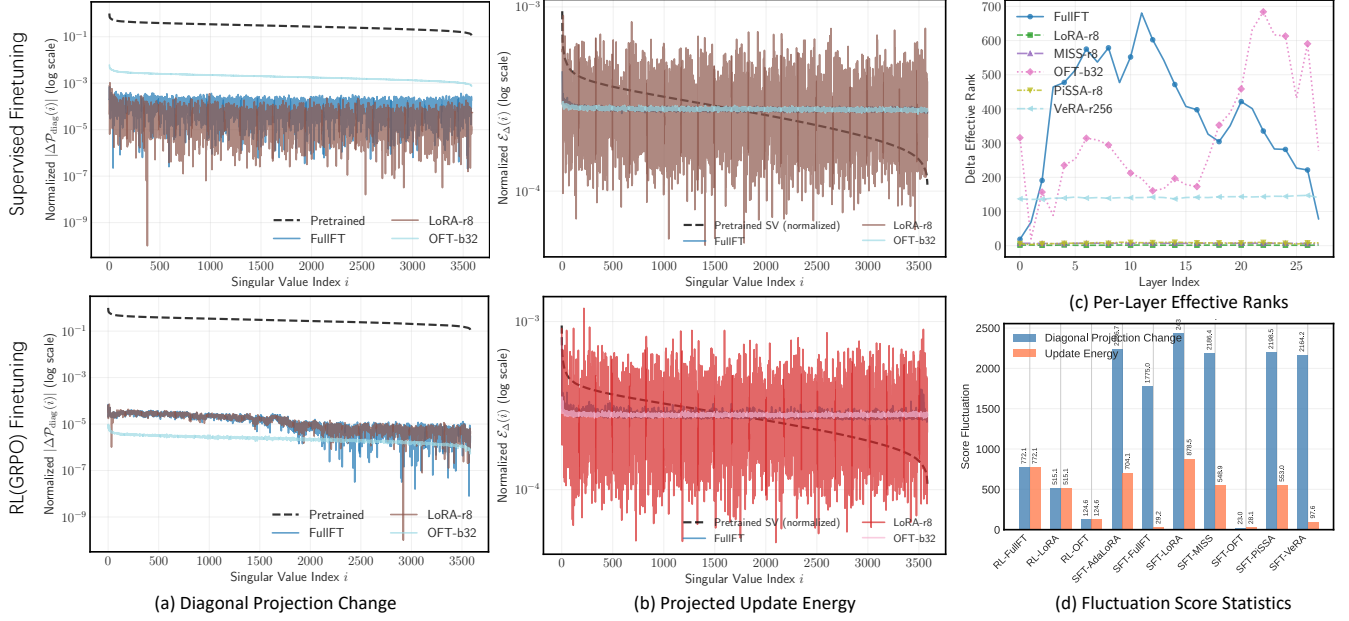
*Fixed update directions.* VeRA[30] freezes update directions using randomly initialized high-rank matrices. This typically yields lower plasticity (limited ability to align with task-relevant directions), but also relatively low interference because the updates are closer to orthogonal to the dominant pre-trained components.

*Spectrum Preservation of Orthogonal Finetuning.* OFT[54] shows **Mode C**-like behavior in **both SFT and RLVR**. This is consistent with its spectrum-preserving nature: applying an (approximately) orthonormal transformation primarily rotates weight vectors without strongly distorting their spectral geometry, providing a stability-inducing “geometric barrier.” As a result, OFT can achieve strong adaptation in both settings, and remains more robust to SFT over-shoot, mitigating forgetting.

*Understanding OFT Forgetting through singular vector alignment (SVA).* Unlike additive updates, OFT is not naturally characterized by  $\Delta W$ . We therefore analyze OFT through *Singular Vector Alignment* (SVA). Let  $W_{\text{pre}} = U_0 \Sigma_0 V_0^\top$ . OFT approximately induces a rotation on the right singular vectors,

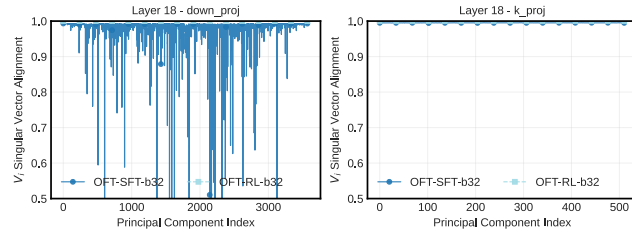
$$V^* \approx V_0 R, \quad R^\top R = I, \quad (3)$$

which yields  $W^* \approx U_0 \Sigma_0 (V_0 R)^\top$ . We compare the cosine similarity between  $V_0$  and  $V^*$  at the same singular-value index to measure rotation intensity. As shown in Figure 3, in RL training OFT exhibits



**Figure 2: Dual-view spectral analysis. Left: effective rank of weight updates. Mid & right: distributions of projected spectrum changes (retention/adaptation views) and their spectral smoothness.**

nearly uniform rotation across components, suggesting a global rigid transformation of layer weights. However, in SFT training, spikes in SVA curves demonstrate there are localized rotations on a small mounts of components in specific layers. This further explains that, though mitigated by OFT, forgetting could still occur during SFT post-training.



**Figure 3: Singular vector alignment (SVA) analysis of OFT updates in different layers.**

## 5 Interpolation Trade-off Curves

Model merging methods[47, 71, 73] expose cross-task trade-offs by combining weights from different task-specialized models. We adopt this idea to analyze the plasticity-stability frontier in SFT alignment. We do not treat a fine-tuned checkpoint as a single operating point. Instead, we evaluate a continuum of interpolated models. This lets us trace a Target-General trade-off curve.

*Task arithmetic.* [21] Let  $W_0$  be the pre-trained weights and  $W^*$  be the aligned weights after SFT on the target domain. The canonical interpolation (“task arithmetic”) is

$$W(\alpha) = W_0 + \alpha (W^* - W_0), \quad \alpha \in [0, 1]. \quad (4)$$

Here,  $\alpha = 0$  recovers the base model and  $\alpha = 1$  recovers the fully adapted model. By sweeping  $\alpha$  and evaluating both target performance and general ability preservation, we obtain an explicit trade-off curve and can identify a “sweet spot” (often at  $\alpha < 1$ ) where gains on the target domain are retained while forgetting is mitigated.

For additive PEFT methods(LoRA/AdaLoRA/MiSS), interpolation corresponds to scaling the learned update  $\Delta W$  with  $\alpha$ . For OFT, we interpolate by scaling the skew-symmetric generator  $Q$  in its Cayley-parameterized rotation. Please check the details and derivations of PEFT interpolation in the Appendix subsection A.2.

## 5.1 Results: The Over-Shoot SFT Adaption

*Interpolation reveals a “free lunch” region.* As shown in Figure 1, we construct interpolation-based trade-off curves by sweeping  $\alpha \in \{0.1, 0.2, \dots, 0.9\}$ . We focus on Qwen2.5-7B and report results on two target domains (math and medical). For each PEFT method, we include two configurations with comparable parameter budgets (shown as solid vs. dashed curves).

Across a wide range of methods, we observe a consistent pattern. This includes additive updates (e.g., LoRA-family) and orthogonal updates (OFT). Back-interpolating from the final checkpoint ( $\alpha = 1$ ) often yields a markedly better stability point. In particular, as shown in Figure 4, many methods achieve an almost “lossless” improvement around  $\alpha \approx 0.2$  to  $0.3$ . Target-domain performance remains close to (or even slightly better than) the  $\alpha = 1$  checkpoint. Meanwhile, general abilities are substantially recovered. This indicates that, in our SFT alignment setting, the final trained checkpoint frequently *overshoots* the best plasticity-stability trade-off. A simple

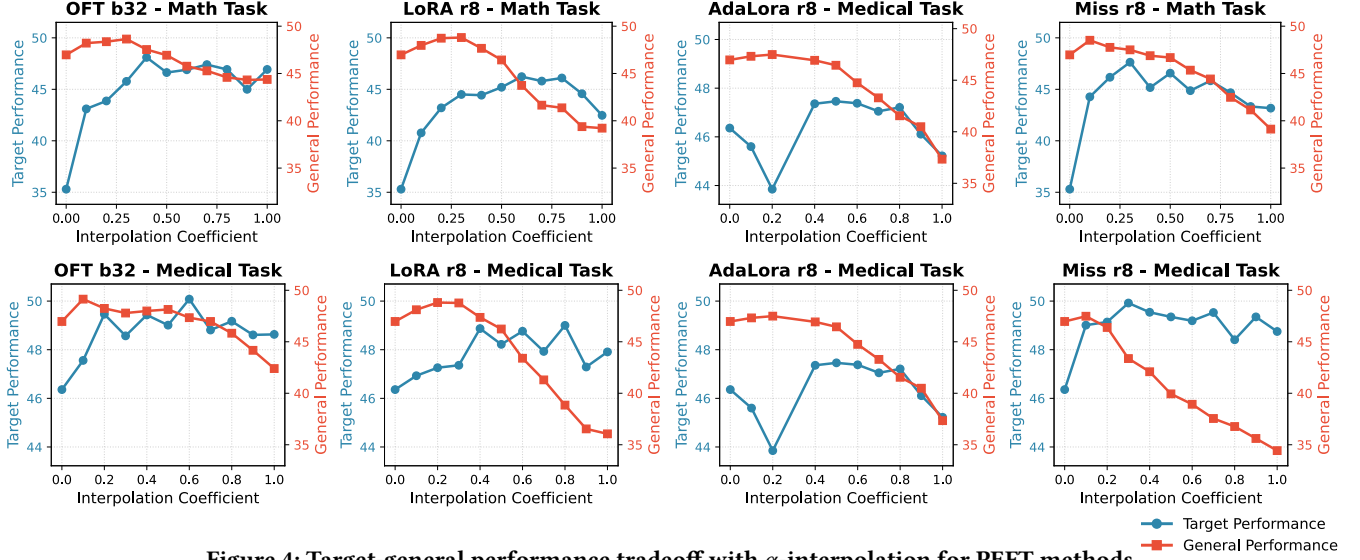
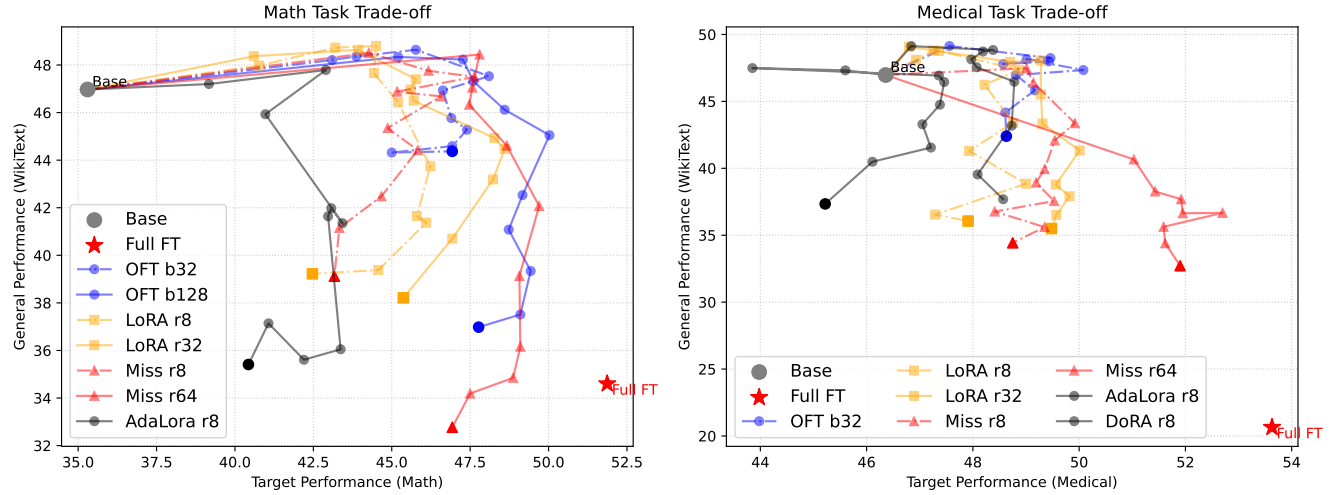
Figure 4: Target-general performance tradeoff with  $\alpha$ -interpolation for PEFT methods

Figure 5: Interpolation trade-off curves for PEFT methods. OFT achieves better "sweet spot" under the same parameter scale.

one-dimensional interpolation can move the model back to a better operating region.

*OFT is Pareto-optimal under interpolation.* Overall, OFT yields the most favorable Pareto frontier: for both domains and for both parameter scales, its interpolation curves lie closer to the upper-right corner, achieving stronger target gains at a smaller stability cost than additive PEFT baselines.

*Optimization vs. interpolation.* We further compare two trajectories. One is the path traced by optimization checkpoints. The other is the interpolation curve between  $W_0$  and  $W^*$ .

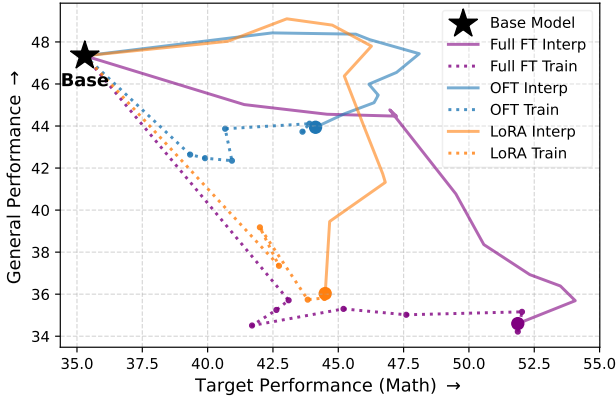
These two trajectories are not only misaligned. They also have opposite curvature. The optimization trajectory is approximately concave in our trade-off plane. General ability typically drops first.

Target performance improves later. In contrast, the interpolation trajectory is convex. As  $\alpha$  increases, target performance often improves first. General ability then drops sharply.

This pattern suggests that the overshoot phenomenon is not a simple overfitting issue that can be fixed by early stopping or regularization. Instead, interpolation provides a practical trade-off knob during PEFT training. It also serves as a diagnostic tool to reveal where stability is lost.

## 5.2 iOFT and Alternative Trade-off Methods

To further analyze the overshoot and the "free lunch" phenomenon, we go beyond a single global interpolation coefficient and explore geometry-aware adjustment and structured pruning on OFT updates. **iOFT is the main focus of this section**, while



**Figure 6: Comparing training trajectories and interpolation trajectories. The misalignment further reveal the overshoot issue of SFT.**

the remaining strategies are included as comparisons/alternatives. Figure 8 and Table 2 summarize these alternatives.

**iOFT: interpolated OFT via layer-wise geometric adjustment.** Empirically, we observe that OFT exhibits a pronounced layer profile of rotation strength: later layers rotate much more than early layers, and this gap widens during training. Motivated by this intrinsic update structure, we introduce **iOFT** (interpolated OFT), which rebalances the update by rescaling the generator  $Q_\ell$  per layer so that the effective rotation strengths become more uniform. Concretely, iOFT instantiates two simple reference rules: **SafeRho** matches each layer to the average rotation strength of the first five layers, while **MinRho** matches each layer to the minimum rotation strength across layers. Because iOFT leverages the model’s own weight-update geometry, it exhibits more robust trade-off improvements across settings: it can consistently recover general abilities while preserving target gains without requiring an exhaustive sweep over a global scaling factor.

In Figure 2, we show that forgetting is often associated with *spiky* and *incoherent* spectral deviations, reflected by high fluctuations in both the retention view ( $|\Delta P_{\text{diag}}(i)|$ ) and the adaptation view ( $E_\Delta(i)$ ). A sharp layer-wise spike in rotation strength  $\rho_\ell$  implies that a small set of layers (typically later layers) undergo much stronger geometric re-allocation than others, which can translate into uneven, layer-localized spectral changes and thus increase the risk of destructive interference. By flattening the  $\rho_\ell$  profile through layer-wise rescaling of  $Q_\ell$ , iOFT can reduce such layer-induced spectral spikes, providing a mechanism-level rationale for its improved stability.

**Global scaling requires tuning  $\alpha$  across training settings.** In contrast, when we apply a uniform scaling to the entire OFT update (analogous to choosing a global interpolation coefficient  $\alpha$ ), the optimal trade-off is sensitive to the training setting and the overall update magnitude. In practice, selecting the best global scale typically requires enumerating candidate values and evaluating them to find the best  $\alpha$ .

| Variant                    | Target(Math)   | General(Math) | Target(Med)   | General(Med)  |
|----------------------------|----------------|---------------|---------------|---------------|
| Finetuned                  | 46.93 (+11.63) | 44.37 (-2.60) | 48.63 (+2.27) | 42.40 (-4.57) |
| Uniform scaling 0.8        | 46.93 (+11.63) | 44.82 (-2.15) | 49.17 (+2.81) | 45.76 (-1.21) |
| Uniform scaling 0.4        | 48.10 (+12.80) | 47.53 (+0.56) | 48.83 (+2.47) | 48.15 (+1.18) |
| Adaptive scaling (SafeRho) | 47.17 (+11.87) | 46.69 (-0.28) | 50.01 (+3.65) | 47.61 (+0.64) |
| Adaptive scaling (MinRho)  | 47.83 (+12.53) | 46.86 (-0.11) | 49.76 (+3.41) | 47.79 (+0.82) |
| BlockDrop 0.2              | 46.13 (+10.83) | 45.70 (-1.64) | 48.64 (+2.29) | 46.84 (-0.50) |
| BlockDrop 0.4              | 44.23 (+8.93)  | 47.27 (-0.07) | 49.86 (+3.50) | 47.45 (+0.10) |
| LayerDrop 18-27            | 47.03 (+11.73) | 46.07 (-1.27) | 49.26 (+2.90) | 47.35 (+0.01) |

**Table 2: Ablation of Trade-off Alternatives on OFT Updates.**

**Block/layer dropping as coarse alternatives.** We also explore pruning OFT updates at a coarser granularity. We apply random block dropping (BlockDrop) to the diagonal block rotation matrices, and we remove updates in selected layers (LayerDrop). While these methods are generally less effective than iOFT in improving the target-general frontier, they can still substantially mitigate forgetting and often retain a large portion of the target improvements.

**Implications.** Taken together, these results suggest that overshoot is closely tied to the geometry and distribution of the learned update across layers. Our exploration provides additional motivation and practical references for future work on improving PEFT stability-plasticity trade-offs by exploiting parameter geometry, both within layers (intra-layer) and across layers (inter-layer), rather than relying solely on global scaling and post-hoc  $\alpha$  selection.

## 6 Concluding Remarks

We present **PEFT-Arena**, a novel benchmark to evaluate PEFT jointly on **plasticity** (target-domain gains) and **stability** (general ability preservation) by pairing domain-specific adaptation (math and medical) with general capability retention. Within PEFT-Arena, **OFT** with spectrum-preserving update consistently achieves a stronger Pareto frontier than additive low-rank baselines under comparable parameter budgets.

To connect these empirical trade-offs to internal mechanisms, we develop a **dual-axes (dual-view) spectral analysis** of weight updates, separating **retention** (alignment with the pretrained singular basis) from **adaptation** (how update energy is distributed). This analysis suggests two key findings: (i) a **geometric barrier**, where structure-preserving, rotation-like updates maintain smoother and more coherent spectral deviations and thus reduce destructive drift; and (ii) a tendency of many constrained additive parameterizations to exhibit **spiky** and **incoherent** spectral changes under SFT, which correlates with increased forgetting.

Finally, we identify a robust **SFT overshoot** phenomenon: the final adapted checkpoint ( $\alpha = 1$ ) often lies beyond the best stability-plasticity operating point. To make this explicit, we use **interpolation** between the base and adapted weights to obtain **Interpolation Trade-off Curves**, which frequently uncover a “free-lunch” region that recovers general abilities with little or no target loss. Building on this observation, we propose **iOFT**, a simple empirical, geometry-aware variant that rebalances OFT updates via layer-wise adjustment, providing a practical robust trade-off knob beyond sweeping a global scaling coefficient.

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## A Appendix

### A.1 Evaluation details

*Evaluation settings for tasks of different domains.* **Math.** Each math problem is evaluated by average accuracy@16, with a maximum response length of 8192 tokens (consistent with training) and temperature  $T = 0.6$ . **Medical.** We use average accuracy as the primary metric with temperature  $T = 0.0$ . **General.** We follow the evaluation configuration in OpenCompass<sup>1</sup> with context length 1024, temperature  $T = 0.0$ , and one sample per query.

*Training details.* **Compute.** All experiments are conducted on 8× NVIDIA H100 80GB GPUs. **SFT.** We use an effective batch size of 256, a maximum response length of 8192 tokens, and train for 4 epochs. Full fine-tuning uses a learning rate of  $5 \times 10^{-5}$ , while all PEFT methods use  $2 \times 10^{-4}$ . We adopt a cosine decay learning-rate scheduler. **RL.** We use an effective batch size of 256, a mini-batch size of 64, and a group size of 8. The maximum generation length is set to 8192 tokens, consistent with SFT and evaluation. Full fine-tuning uses a learning rate of  $10^{-6}$ , while all PEFT methods use  $10^{-5}$ . We train for 200 steps for all RL experiments.

### A.2 Interpolation implementation details

*Additive-update PEFT (LoRA/AdaLoRA/MiSS).* For PEFT methods with an additive parameterization,

$$W^* = W_0 + \Delta W, \quad W(\alpha) = W_0 + \alpha \Delta W. \quad (5)$$

Interpolation is implemented by scaling the learned update  $\Delta W$ . For LoRA,  $\Delta W = s BA$  (with method-dependent scale  $s$ , e.g.,  $s = \frac{\alpha \text{LoRA}}{r}$ ). A convenient implementation is to scale both factors,

$$\Delta W(\alpha) = s(\sqrt{\alpha}B)(\sqrt{\alpha}A) = \alpha s BA, \quad (6)$$

which preserves the product structure and avoids excessively scaling a single factor.

*OFT: interpolating rotation by scaling the generator.* Using the Cayley form, the rotation is parameterized as

$$R(Q) = (I + Q)(I - Q)^{-1}, \quad (7)$$

where  $Q$  is the skew-symmetric generator. We interpolate OFT by scaling its generator,

$$R(\alpha) \triangleq R(\alpha Q) = (I + \alpha Q)(I - \alpha Q)^{-1}, \quad \alpha \in [0, 1]. \quad (8)$$

For small angles, the rotation angle  $\theta$  satisfies  $\|Q\| \approx \tan(\theta/2) \approx \theta/2$ , hence  $\theta$  is approximately linear in  $\|Q\|$ . Following prior geometric diagnostics, we define the layer-wise rotation strength

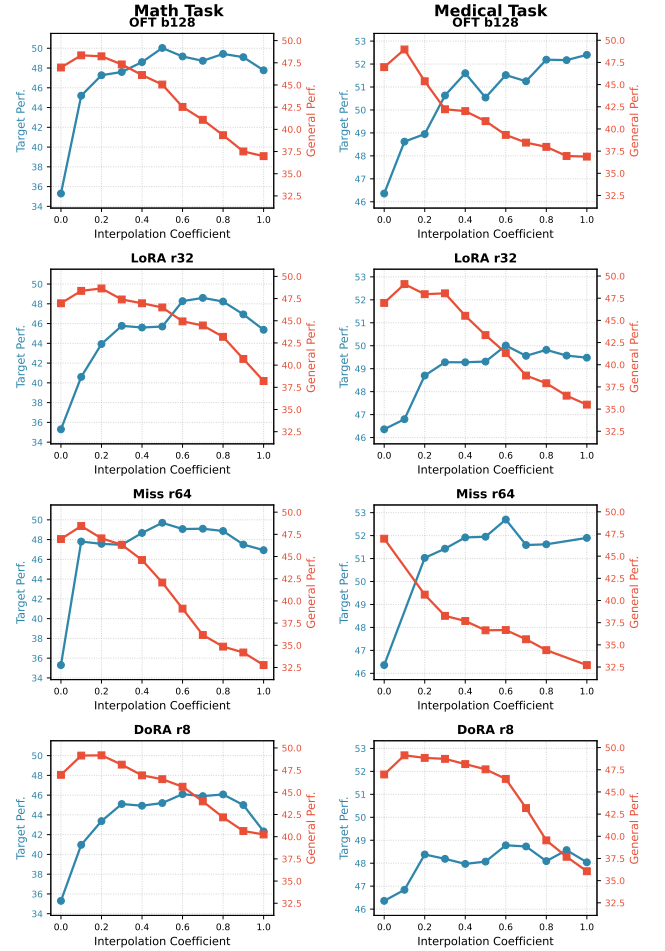
$$\rho_\ell \triangleq 1 - \cos(\theta_\ell). \quad (9)$$

Using  $\cos \theta \approx 1 - \theta^2/2$  gives  $\rho_\ell \approx \theta_\ell^2/2$ , and therefore  $\rho_\ell \propto \|Q\|^2$ . Consequently, to scale the rotation strength by a factor of  $k$  in the small-angle regime, we scale the generator by  $\sqrt{k}$ :  $Q' = \sqrt{k} Q$ .

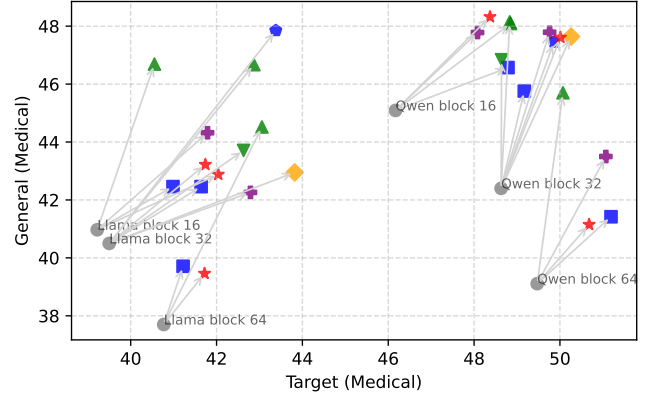
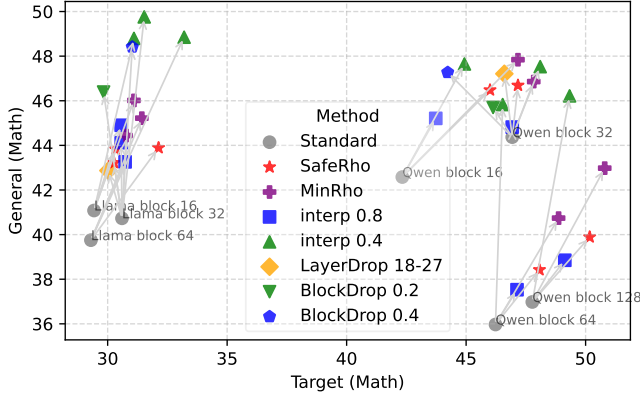
### A.3 Additional Results

Here we provide additional results for analysis in section 4 Spectral Analysis (Figure 9) and section 5 Interpolation Trade-off (Figure 8, subsection A.3) to comprehensively demonstrate our experimental observations.

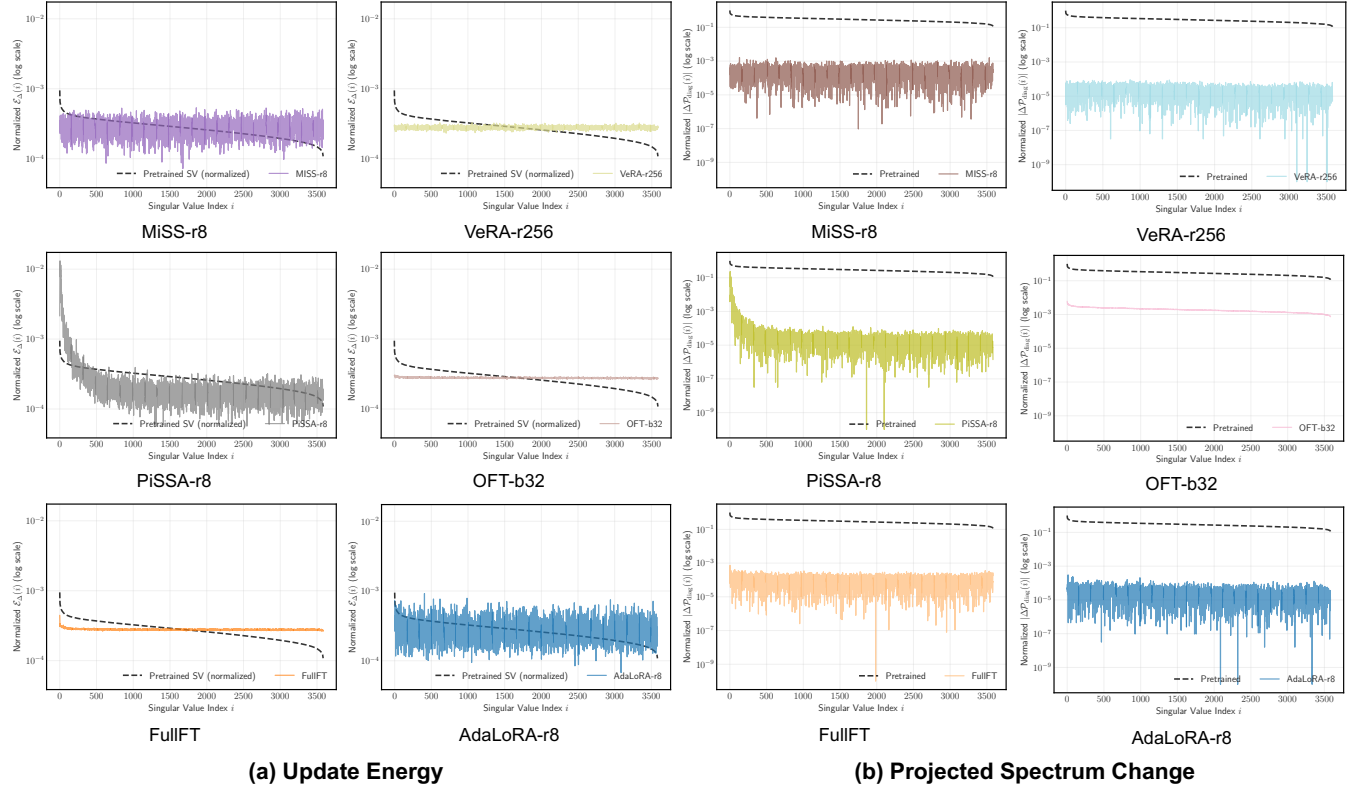
<sup>1</sup><https://github.com/open-compass/opencompass>



**Figure 7: (Additional) Target-General Performance Tradeoff with  $\alpha$ -Interpolation for PEFT Methods.**



**Figure 8: Trade-off alternatives achieve consistent improvement on OFT adapters of different sizes. This figure complements Table 2 by visualizing how different scaling/pruning alternatives behave across adapter sizes.**



**Figure 9: Additional visualization for spectrum analysis in SFT Training. Including distributions of (a) project update energy and (b) diagonally projected spectrum changes on pretrained basis.**